



# PROJECT PROPOSAL: INNOVYX BOQ PROGRAM

## Web-Based Bill of Quantities (BOQ) Data Processing Platform

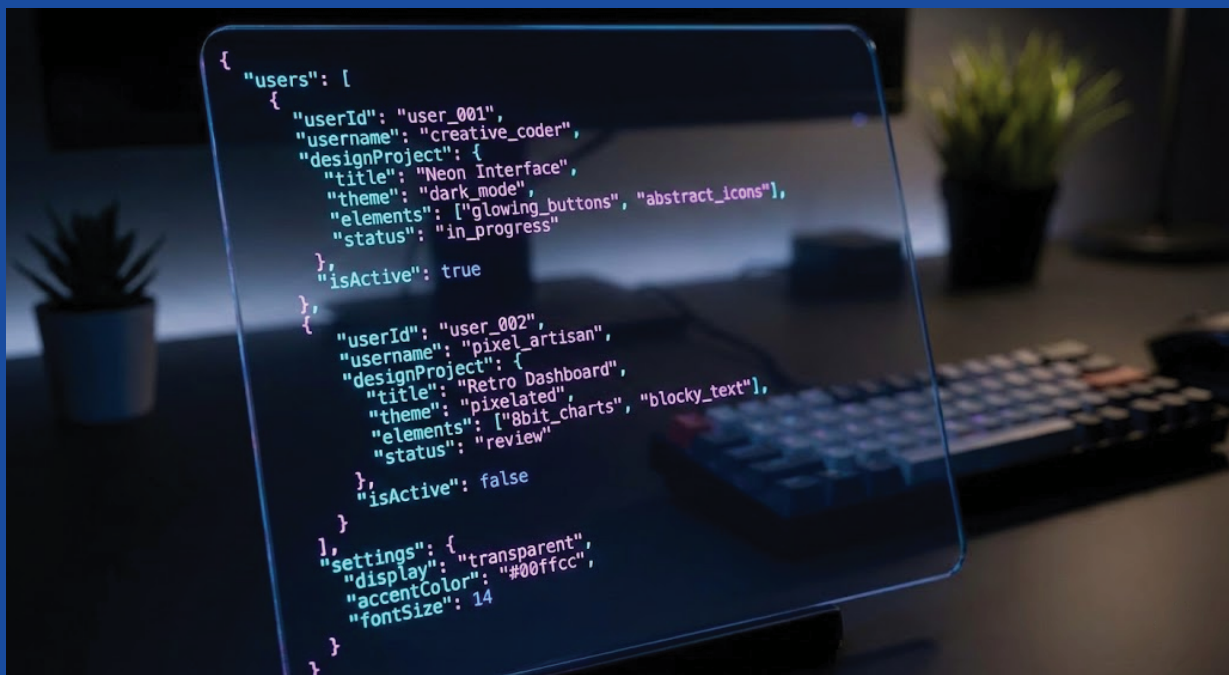
|                |                     |              |                       |
|----------------|---------------------|--------------|-----------------------|
| Project Name : | Innovyx BOQ Program | Prepared By: | INNOVYX TECH LABS LLP |
| Version:       | 1.0                 | Date:        | December 1, 2025      |



## 1. Executive Summary

This proposal outlines the development of a modern, web-based data processing and spreadsheet management platform designed to streamline the Bill of Quantities (BOQ) workflow for the Innovyx Program. The solution is built to leverage JSON data from external systems (like CPQ), transform it into an interactive, Excel-like environment for modification and pricing, and manage critical revision controls.

The project is segmented into two distinct phases to ensure a focused, modular approach to delivery.



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## 2. Project Objective

The primary objective is to develop a robust, web-based platform that integrates the flexibility of JSON data with the analytical power of an Excel-like interface.

### Key Goals include:

- **Seamless JSON-to-Spreadsheet Transformation:** Provide users with the ability to upload or fetch JSON data and automatically convert it into a structured, editable spreadsheet format.
- **Advanced In-Browser Functionality:** Deliver a high-performance spreadsheet engine supporting over 400 Excel functions, real-time formula evaluation, and advanced lookup operations.
- **Comprehensive BOQ Support:** Enable users to upload, manage, and calculate BOQ data, supporting industry workflows for estimation and project costing.
- **Secure Access Control:** Implement a robust Role-Based Access Control (RBAC) system, ensuring users only access datasets assigned to them.



### 3. Technology Stack & Architecture

The platform will be built on a modern, scalable architecture:

- **Back-end:** Python
- **Frontend:** React/Next.js
- **Spreadsheet Engine:** Hyper-Formula (to ensure local, high-speed calculation support for over 400 Excel functions)



## 4. Detailed Scope of Work

The scope is divided into two steps, mirroring the process flow discussed and highlighted in the attached flowchart.

### Step 1: Initial BOQ Generation and Editing (The Red Box - MVP Scope)

This phase covers the foundational work required for data ingestion, visualization, and basic editing, as represented by the initial stages of the flowchart.

| Feature Category               | Scope Detail (Mapping to Documents)                                                                                                                | Source Reference                                              |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| 1. JSON Ingestion              | Connect with the CPQ/ SystemSummary JSON file and automatically transform it into a spreadsheet grid.                                              | JSON-to-Spreadsheet Transformation and JSON Upload Options.   |
| 2. BOQ Generation              | Generate the Master BOQ and Modified BOQ based on the selected BOQ format (implied structure) using the JSON data.                                 | BOQ Support and JSON Conversion.                              |
| 3. Online Editing Interface    | Provide an online, interactive, Excel-like spreadsheet editor for real-time manipulation of the BOQ data.                                          | Spreadsheet Editor, Cell Editing & Formula Bar.               |
| 4. Structural Management       | Allow the user to <b>add/delete rows and columns</b> within the BOQ structure as required.                                                         | Row & Column Management.                                      |
| 5. Cell Validations & Formulas | Implement cell validations (as required) and integrate the formula engine to support BOQ calculations and general Excel functions within the grid. | Excel Formula Support and BOQ Calculations (Quantity x Rate). |
| 6. Data Security (Foundation)  | Implement the secure Login and Authentication module to restrict access to the assigned datasets.                                                  | User Login & Access Control.                                  |
| 7. Data Output                 | Allow the export of the Modified BOQ data in both <b>JSON</b> and <b>Excel (.xlsx)</b> formats.                                                    | Exporting Data.                                               |

## Step 2: Pricing, Database Integration, and Revision Control

This phase completes the BOQ process by integrating external pricing data, enabling engineering modifications, and establishing an auditable revision workflow.

| Feature Category                         | Scope Detail (Mapping to Documents)                                                                                                                                                                      | Source Reference                                          |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| <b>1. Pricing Database Integration</b>   | Design and implement the functionality to store the priced <b>BOQ from CRM into a Price Database by Project</b> . This requires API definition (Future Scope) or direct database connection.             | Database Integration (implied by "Store the priced BOQ"). |
| <b>2. Modification Workflow</b>          | Implement the workflow allowing the <b>Engineering Department to modify the BOQ</b> (the "Revised BOQ" in the flowchart) while maintaining access control and permissions.                               | Access Restrictions and Structural Editing                |
| <b>3. Price Matching/ Fetching</b>       | Functionality to fetch existing item prices from the Price Database and automatically populate matching items in the Revised BOQ.                                                                        | Price Database connection (implied).                      |
| <b>4. Missing Item Identification</b>    | Identify the list of items that are missing prices and require integration with <b>CRM</b> (requires IT/API input as per request).                                                                       | Missing Item Identification (New functionality).          |
| <b>5. Revision Control &amp; History</b> | Implement a <b>Release/ Revision Control</b> mechanism for the Modified BOQ, allowing the system to <b>maintain revision control for priced BOQs</b> and update the price database in a continuous loop. | Revision Management.                                      |
| <b>6. Audit Trail</b>                    | Maintain a complete audit trail (tracking user, timestamp, previous/new value) for all edits and structural changes made by the Engineering Department.                                                  | Change Tracking and Audit Logs.                           |

## 5. Next Steps and Quotation

Based on the detailed scope breakdown above, please find the commercial quote for the Innovyx BOQ Program attached.

We are ready to align on a kick-off date upon receipt of the Purchase Order for Step 1 (the MVP scope), which allows us to deliver the core BOQ editing functionality immediately.

| Phase  | Description                                                                                              | Estimated Delivery | Cost (Placeholder) |
|--------|----------------------------------------------------------------------------------------------------------|--------------------|--------------------|
| Step 1 | Initial BOQ Generation, JSON Ingestion, Full Excel-like Editor, and Data Export.                         | 350-380 hrs        | [Insert Price]     |
| Step 2 | Pricing Database Integration, Engineering Workflow, Automated Price Fetching, and Full Revision Control. | 375-400 hrs        | [Insert Price]     |

We look forward to partnering with you on this critical project

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